COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
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ACTIVITY NO. 1.1 **Basic C++ Program**

1. PROCEDURE OUTPUT

Sections	**Answer**
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	`#include <iostream>`
	
	`// FUNCTION DECLARATIONS`
	
	`// ILO A DECLARATION`
	`int sumOfTwoNum(int fNum, int sNum); // Number 1`
	
	`bool isGreater(int a, int b); // Number 2`
	
	`bool displayLogicalOperations(bool c, bool d); // Number 3`

Class Declaration and Method Definition Section	`// CLASS`
	`class Triangle {`
	`private:`
	`double totalAngle, angleA, angleB, angleC;`
	`public:`
	`Triangle(double A, double B, double C);`
	`void setAngles(double A, double B, double C);`
	`const bool validateTriangle();`
	`};`

```cpp
// ILO B
Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}


void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}


const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

Main Function

```cpp
// MAIN PROGRAM

int main () {

    // ILO A #1

    std::cout << "The sum of all numbers is : " << sumOfTwoNum(10,22) << std::endl;

    // ILO A #2
    int x = 500;
    int y = 400;

   if (isGreater(x, y)) {
        std::cout << "x is greater than y" << std::endl;
    } else {
        std::cout << "x is not greater than y" << std::endl;
    }

    // ILO A #3

    bool success = displayLogicalOperations(true, false);

    std::cout << "Function execution status: " << std::boolalpha << success << std::endl;

    // ILO B
    Triangle set1(40,50,110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid Triangle \n.";
    }
    else{
        std::cout << "The shape is NOT a valid Triangle \n.";
    }

    return 0;
}
```

Method Definition

```cpp
// ILO A #1
int sumOfTwoNum(int fNum, int sNum){


    int sum = fNum + sNum;
    return sum;
}
 // ILO A #2
bool isGreater(int a, int b){
    return a > b;
}
```

```
127  // ILO B
128  Triangle::Triangle(double A, double B, double C){
129      angleA = A;
130      angleB = B;
131      angleC = C;
132      totalAngle = A + B + C;
133  }
134
135
136  void Triangle::setAngles(double A, double B, double C){
137      angleA = A;
138      angleB = B;
139      angleC = C;
140      totalAngle = A+B+C;
141  }
142
143
144  const bool Triangle::validateTriangle(){
145      return (totalAngle <= 180);
146  }
```

```cpp
// ILO A #3
bool displayLogicalOperations(bool c, bool d) {
    // std::boolalpha formats outputs as "true" or "false" instead of "1" or "0"
    std::cout << std::boolalpha;

    std::cout << "Logical Operations Table " << std::endl;
    std::cout << "Inputs: A = " << c << ", B = " << d << std::endl;

    // 1. NOT
    std::cout << "NOT A      : " << (!c) << std::endl;
    std::cout << "NOT B      : " << (!d) << std::endl;

    // 2. AND
    std::cout << "LOGICAL AND : " << (c && d) << std::endl;

    // 3. OR
    std::cout << "LOGICAL OR : " << (c || d) << std::endl;

    // 4. XOR
    std::cout << "LOGICAL XOR : " << (c ^ d) << std::endl;


    // 5. NAND
    std::cout << "LOGICAL NAND: " << !(c && d) << std::endl;


    // 6. NOR
    std::cout << "LOGICAL NOR: " << !(c || d) << std::endl;


    // 7. XNOR
    std::cout << "LOGICAL XNOR: " << !(c ^ d) << std::endl;

    return true;
}
```

Table 1-1. C++ Structure Code for Answer

```cpp
// ILO B
Triangle set1(40,50,110);
if(set1.validateTriangle()){
    std::cout << "The shape is a valid Triangle \n.";
}
else{
    std::cout << "The shape is NOT a valid Triangle \n.";
}
```

```
The shape is NOT a valid Triangle
```

The program displayed "The shape is NOT a valid Triangle." because the angles 40°, 50°, and 110° have a total

of 200°, which is greater than 180°.. The activity helped me understand how constructors, objects, and member functions work together to validate data in C++..

Table 1-2. ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1.Create a C++ program to swap the two numbers in different variables.
2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
3. Create a C++ program that has a function that will calculate the distance between two points.
4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or ‘others.’

HOA1Supplementary.cpp > main()

```cpp
#include <iostream>
#include <cmath>
#include <string>

// Task 1 Function Declaration
void numSwap();

// Task 2 Function Declaration
double kelvinToFahrenheit(double kelvin);

// Task 3 Function Declaration
void calculateDistance();
double distanceCalculator(double x1, double x2, double y1, double y2);

// TASK 4 Class

class Triangle {
private:
    double sideA, sideB, sideC;
    double angleA, angleB, angleC;
    double totalAngle;

public:
    Triangle(double sA, double sB, double sC,
             double A, double B, double C);

    void setTriangle(double sA, double sB, double sC,
                     double A, double B, double C);

    bool validateTriangle();

    double getArea();

    double getPerimeter();

    std::string triangleType();
};
```

```cpp

// MAIN PROGRAM
int main() {

    double kelvin;

    Triangle tri(5, 6, 7, 40, 50, 90);

    std::cout << "Task #1 Swapping two numbers in different variables." << std::endl;
    numSwap();

    std::cout << "\nTask #2 Function converting temperature in Kelvin to Fahrenheit" << std::endl;
    std::cout << "Enter Temperature you want to convert in Kelvin: ";
    std::cin >> kelvin;

    std::cout << "\nTemperature converted to Fahrenheit: "
              << kelvinToFahrenheit(kelvin) << " Fahrenheit" << std::endl;

    std::cout << "\nTask #3 Calculating Distance Between Two Points" << std::endl;
    calculateDistance();

    std::cout << "\nTask #4 Modifying Triangle Class and getting Area, Perimeter, Triangle Type Dete

    if (tri.validateTriangle()) {
        std::cout << "The triangle is valid.\n";
        std::cout << "Perimeter : " << tri.getPerimeter() << std::endl;
        std::cout << "Area      : " << tri.getArea() << std::endl;
        std::cout << "Type      : " << tri.triangleType() << std::endl;
    }
    else {
        std::cout << "The triangle is NOT valid.\n";
    }

    return 0;
}
```

```cpp
// FUNCTION DEFINITIONS

// Task 1
void numSwap() {

    int a, b, temp;

    std::cout << "Input Values for A and B\n";
    std::cout << "A = ";
    std::cin >> a;
    std::cout << "B = ";
    std::cin >> b;

    std::cout << "\nCurrent Values of A and B Before Swapping\n";
    std::cout << "A = " << a << std::endl;
    std::cout << "B = " << b << std::endl;

    temp = a;
    a = b;
    b = temp;

    std::cout << "\nValues after swapped" << std::endl;
    std::cout << "A = " << a << std::endl;
    std::cout << "B = " << b << std::endl;
}

// Task 2
double kelvinToFahrenheit(double kelvin) {

    return (kelvin - 273.15) * 9 / 5 + 32;

}
```

```cpp
// Task 3
void calculateDistance() {

    double x1, x2, y1, y2;

    std::cout << "Enter x1: ";
    std::cin >> x1;

    std::cout << "Enter y1: ";
    std::cin >> y1;

    std::cout << "Enter x2: ";
    std::cin >> x2;

    std::cout << "Enter y2: ";
    std::cin >> y2;

    std::cout << "\nDistance Calculated from your Inputs:" << std::endl;
    std::cout << "X1 : " << x1 << std::endl;
    std::cout << "Y1 : " << y1 << std::endl;
    std::cout << "X2 : " << x2 << std::endl;
    std::cout << "Y2 : " << y2 << std::endl;

    std::cout << "Distance is = " << distanceCalculator(x1, x2, y1, y2) << std::endl;
}

double distanceCalculator(double x1, double x2, double y1, double y2) {

    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

}
```

```cpp
// Task 4 Function Definitions

// Constructor
Triangle::Triangle(double sA, double sB, double sC,
                   double A, double B, double C)
{
    sideA = sA;
    sideB = sB;
    sideC = sC;

    angleA = A;
    angleB = B;
    angleC = C;

    totalAngle = A + B + C;
}

// Setter
void Triangle::setTriangle(double sA, double sB, double sC,
                           double A, double B, double C)
{
    sideA = sA;
    sideB = sB;
    sideC = sC;

    angleA = A;
    angleB = B;
    angleC = C;

    totalAngle = A + B + C;
}
```

```cpp
// Validator
bool Triangle::validateTriangle()
{
    return totalAngle == 180;
}

// Perimeter
double Triangle::getPerimeter()
{
    return sideA + sideB + sideC;
}

// Area
double Triangle::getArea()
{
    double s = getPerimeter() / 2;

    return sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

// Triangle Type Determiner
std::string Triangle::triangleType()
{
    if (angleA < 90 && angleB < 90 && angleC < 90)
        return "Acute-Angled";

    else if (angleA > 90 || angleB > 90 || angleC > 90)
        return "Obtuse-Angled";

    else
        return "Others";
}
```

OUTPUTS

```
Task #1 Swapping two numbers in different variables.
Input Values for A and B
A = 90
B = 09

Current Values of A and B Before Swapping
A = 90
B = 9

Values after swapped
A = 9
B = 90

Task #2 Function converting temperature in Kelvin to Fahrenheit
Enter Temperature you want to convert in Kelvin: 700

Temperature converted to Fahrenheit: 800.33 Fahrenheit

Task #3 Calculating Distance Between Two Points
Enter x1: 1
Enter y1: 2
Enter x2: 3
Enter y2: 4

Distance Calculated from your Inputs:
X1 : 1
Y1 : 2
X2 : 3
Y2 : 4
Distance is = 2.82843

Task #4 Modifying Triangle Class and getting Area, Perimeter, Triangle Type Determiner
The triangle is valid.
Perimeter : 18
Area      : 14.6969
Type      : Others
PS C:\Users\mendo\OneDrive\Documents\C++\output>
```

Summary of lessons learned

During this activity, I realized that I had become too dependent on using “using namespace std;.” Because of this, I sometimes forgot how to properly write the standard namespace, such as std::cout and std::cin. Practicing without relying on using namespace std; helped me become more familiar with the proper C++ syntax and improved my understanding of how namespaces work.

Analysis of the procedure

While doing the procedure, I noticed how organizing the code into function declarations, the main() function, and function definitions made the program much easier to follow. At first, writing the function definitions below main(), I realized it helped me understand how functions are called and reused. I also appreciated how the class grouped related data and functions together, making the code look cleaner and more organized.

Analysis of the supplementary activity

The supplementary activity gave me a chance to apply the concepts instead of just reading about them. The first three tasks helped me become more comfortable with creating and calling functions, while the last task challenged me to work with classes and member functions. Although the Triangle class was the most difficult part, modifying it helped me understand how functions inside a class can perform different tasks while working with the same data.

Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

Overall, I think I performed well in this activity because most of the concepts served as a review of what I learned in Programming Logic and Design (PLD). The only part I found challenging was working with classes since I was more familiar with them in Python during our Object-Oriented Programming lessons. However, it was a good experience to apply the concept of classes in C++, even though the activity only covered the basics of object-oriented programming. The part I enjoyed the most was using function prototypes, defining the functions below the main() function, and then calling them whenever needed. It was satisfying to see each function work correctly, and it showed me how reusable and organized functions can make a program.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Tools Used: Grammarly

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Grammar Correction / Fixing Sentence Flow

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: First, I made a paragraph in my own words and checked if the flow was thoroughly understandable on what I wanted to say in my answers, mainly the analysis.

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